

Math Primer

(5. Taylor Series)

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- ▶ Introduction to Taylor Series
- ▶ Taylor Series of $f(x)$
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- ▶ Finite differences and errors



Brook Taylor
(1685-1731)



Taylor Series

- ▶ **Taylor Series** or **Taylor expansion** of a function is an infinite polynomial whose coefficients are expressed as derivatives at a single point $x = k$.

$$f(x) = a_0 + a_1(x - k) + a_2(x - k)^2 + \dots + a_n(x - k)^n + \dots$$

- ▶ If the derivatives are considered at $x = 0$, it is also called as **Maclaurin** series (a special case of Taylor series)

$$f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n + \dots$$

- ▶ Taylor series is of fundamental importance in all of Quant finance.



Taylor expansion for $f(x)$

- ▶ Suppose we want to expand an infinitely differentiable function $f(x)$ at $x = k$. First, we write the function in the target form

$$f(x) = a_0 + a_1(x - k) + a_2(x - k)^2 + \cdots + a_n(x - k)^n + \cdots$$

Put $x = k$, to obtain $a_0 = f(k)$

- ▶ Differentiate once w.r.t x , we get

$$f'(x) = a_1 + 2a_2(x - k) + \cdots + na_n(x - k)^{n-1} + \cdots$$

Put $x = k$, we get $a_1 = f'(k)$

- ▶ Differentiate n times w.r.t x , we get

$$f^{(n)}(x) = n! a_n + \frac{(n+1)!}{1!} (x - k) + \cdots$$

Put $x = k$, we get $a_n = \frac{f^{(n)}(k)}{n!}$

Taylor expansion for $f(x)$

► Finally, we get this expression

$$f(x) = f(k) + f'(k)(x - k) + \frac{f''(k)}{2!}(x - k)^2 + \dots + \frac{f^{(n)}(k)}{n!}(x - k)^n + \dots$$

$$\text{Or } f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(k)}{n!}(x - k)^n \quad (\text{Taylor Series expansion})$$

► One get' Maclaurin series expansion by setting $k = 0$, which means expanding $f(x)$ at $x = 0$

$$f(x) = f(0) + f'(0)(x) + \frac{f''(0)}{2!}(x)^2 + \dots + \frac{f^{(n)}(0)}{n!}(x)^n + \dots$$

$$\text{Or } f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}(x)^n \quad (\text{Maclaurin Series expansion})$$

Taylor expansion for $f(x)$



Exercise 1 – Write Taylor series expansion for e^x expanding at (i) $x = 0$ and (ii) $x = 2$

Exercise 2 – Write Taylor series expansion for $\ln(1 + x)$ expanding at $x = 0$

Exercise 3 – Write Taylor series expansion for $\sin(x)$ expanding at $x = 0$

Taylor series convergence

- ▶ The Taylor series expansion is often written

$$f(x) = f(k) + f'(k)(x - k) + \frac{f''(k)}{2!}(x - k)^2 + \dots + \frac{f^{(n)}(k)}{n!}(x - k)^n + R_n(x)$$

$\underbrace{\hspace{15em}}$
 $P_n : (\text{Polynomial of degree } n)$

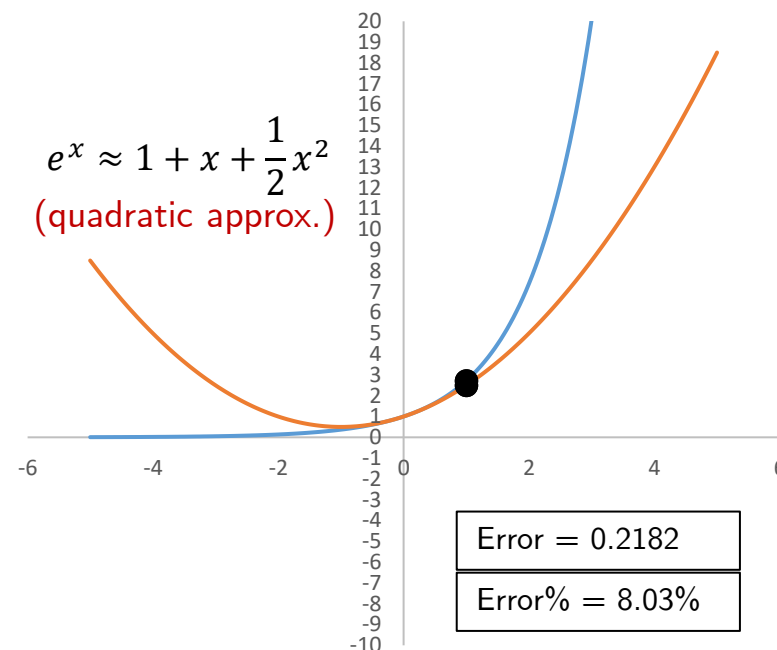
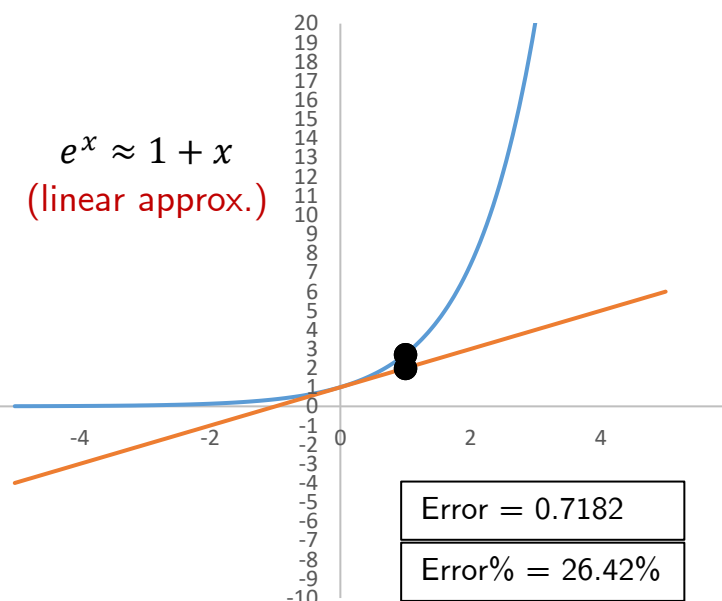
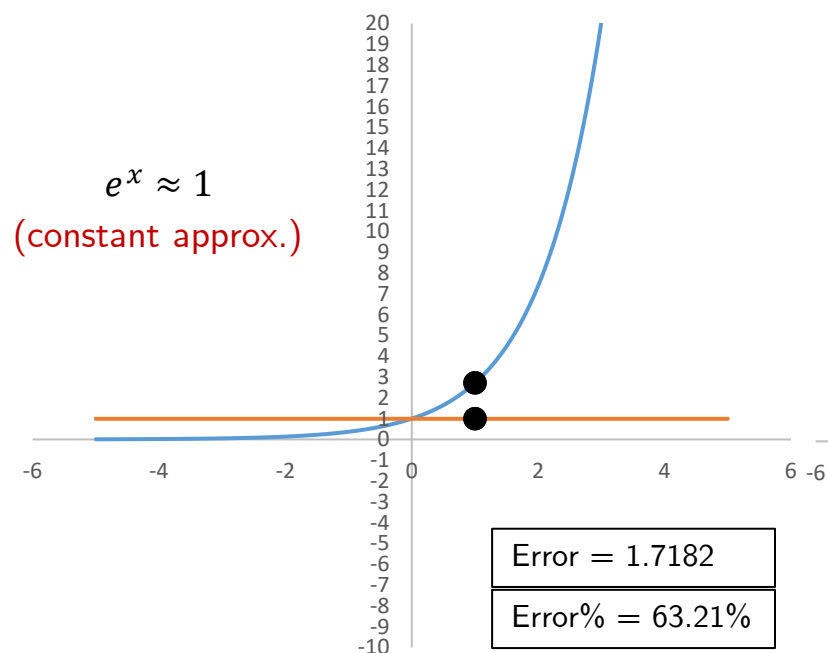
$\underbrace{\hspace{5em}}$
 (remainder)

- ▶ The series is said to converge if $\lim_{n \rightarrow \infty} R_n(x) = 0$
- ▶ For convergent series, the more terms we take, the more accurate the function approximation becomes. In other words, as we go from constant to linear to quadratic approximations, the accuracy increases.

$f(x) \approx f(k)$ (constant approx.)	$f(x) \approx f(k) + f'(k)(x - k)$ (linear approx.)	$f(x) \approx f(k) + f'(k)(x - k) + \frac{f''(k)}{2!}(x - k)^2$ (quadratic approx.)
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Taylor Series Approx. (e^x)

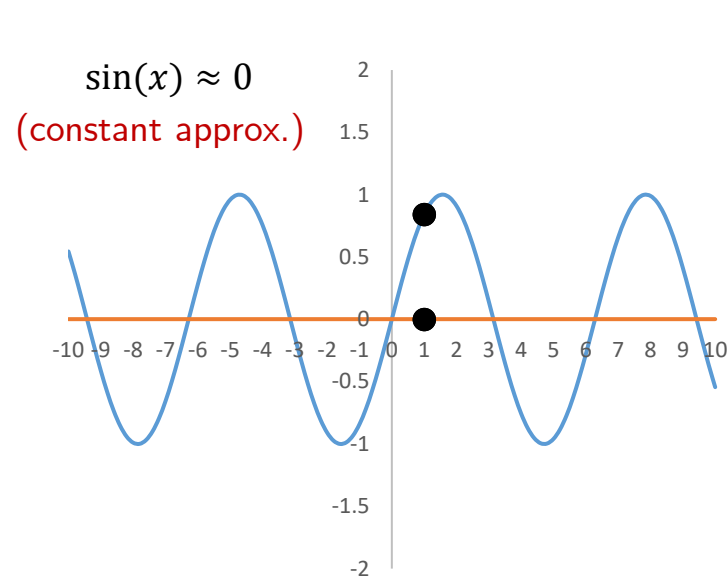
- ▶ Example – Expanding e^x at $x=0$ irrespective of number of terms will always match the function perfectly at $x=0$, but the polynomial approximation will start to deviate as we move away from $x=0$. The more terms we include, the less is the deviation (more accurate)
- ▶ Let's keep visualize the error at $x=1$ as we include more terms



Note that linear approximation is nothing but the tangent to e^x at $x=0$

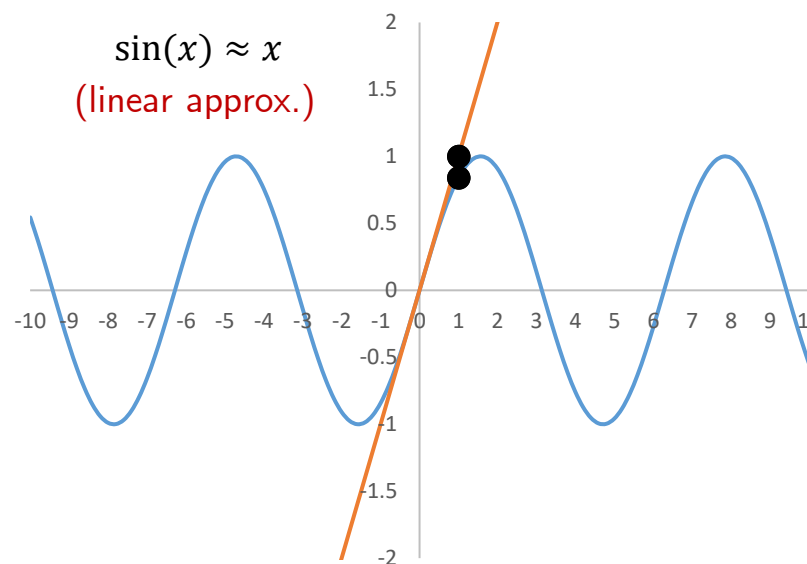
Taylor Series Approx. $\sin(x)$

- ▶ Example – Let's perform the same experiment for $\sin(x)$
- ▶ Let's keep visualize the error at $x=1$ as we include more terms



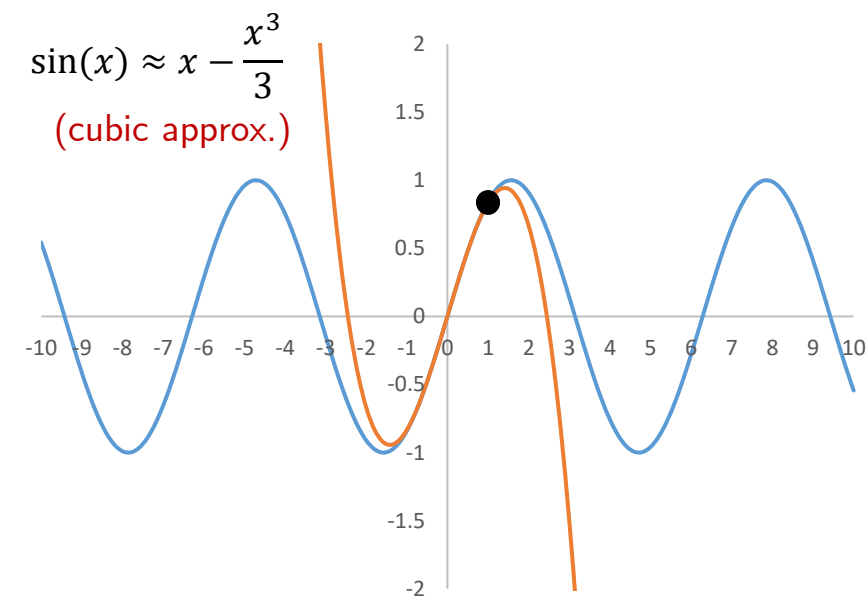
Error = 0.8415

Error% = 100%



Error = -0.1585

Error% = -18.84%



Error = -0.0081

Error% = 0.97%

Taylor Series Approx for change in a function



- ▶ One of the prime usage of Taylor series is to approximate the change in a function when the inputs to the function change. In other words, say $x = x_0$, so $f(x) = f(x_0)$. When x changes from $x_0 + \Delta x$, what is $f(x_0 + \Delta x)$?
- ▶ We can expand $f(x)$ at x_0

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \dots$$

- ▶ Value of function at $x_0 + \Delta x$ is

$$f(x_0 + \Delta x) = f(x_0) + f'(x_0)(\Delta x) + \frac{f''(x_0)}{2!}(\Delta x)^2 + \dots$$

- ▶ We can also write the change in the function

$$\Delta f = f(x_0 + \Delta x) - f(x_0) = f'(x_0)(\Delta x) + \frac{f''(x_0)}{2!}(\Delta x)^2 + \dots$$

Taylor Series Approx for change in a function

- ▶ Truncating the series after the 2nd term, we get the following expression. Note that the first term involves a linear term (proportional change), and the second term involves a quadratic term.

$$\Delta f \approx \underbrace{f'(x_0)(\Delta x)}_{\text{(linear)}} + \underbrace{\frac{1}{2}f''(x_0)(\Delta x)^2}_{\text{(quadratic)}}$$

(First order derivative) (Second order derivative)

- ▶ In practice, we typically work up to the second order term which provides a sufficiently good approximation provided Δx is small. Most of the times, we are even satisfied with the first term only for mathematical convenience.
- ▶ In finance, the left-hand side often represents the change in value of a derivative. In this case, the derivatives represent the sensitivities to risk factors. These sensitivities are called '**delta**' and '**gamma**' respectively. In case of bonds, these get rebranded as **Duration** and **Convexity**.

Taylor Series Approx for change in a function



Exercise 1 – The price of a zero-coupon bond maturing at time T is given as $V = e^{-yT}$ where y is the yield. Approximate the change in bond price up to the second order due to change in yield.

Exercise 2 – The price of a T-coupon bond with face value = F and annual coupon payments @ $c\%$ under flat yield curve is given as

$$V = \frac{Fc}{(1+y)} + \frac{Fc}{(1+y)^2} + \dots + \frac{Fc}{(1+y)^T} + \frac{F}{(1+y)^T}$$

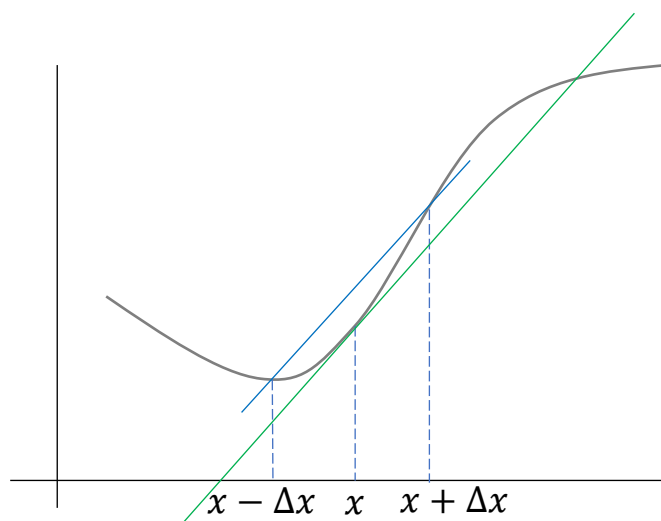
Approximate the change in bond value up to the first order because of change in yield.

Finite Differences and errors

- ▶ In practice, a lot of the time derivatives are approximated using finite differences
- ▶ For first order sensitivity (aka delta), we have three popular choices

(centre delta)

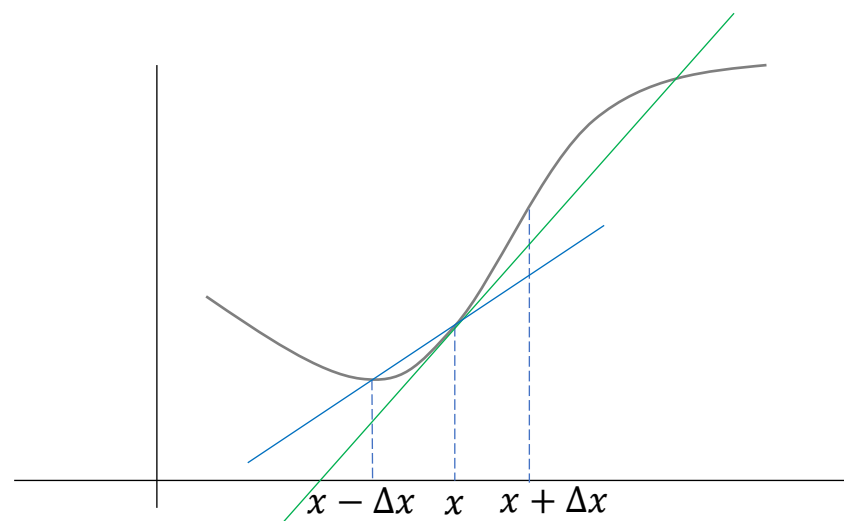
$$\frac{f(x + \Delta x) - f(x - \Delta x)}{2\Delta x}$$



■ $f(x)$ ■ FD ■ True derivative

(backward delta)

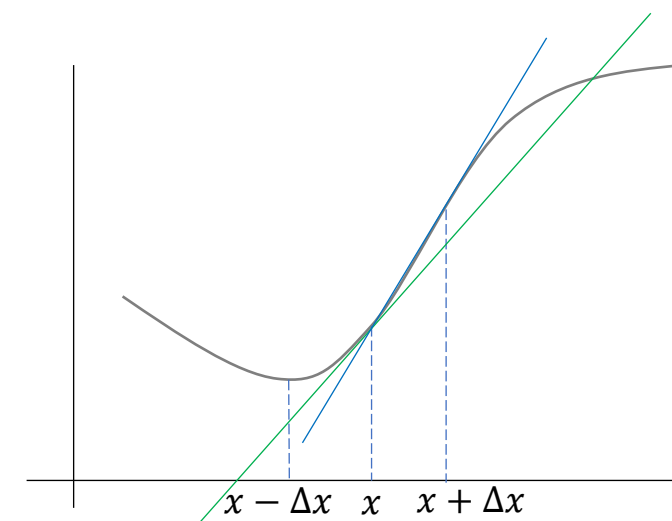
$$\frac{f(x) - f(x - \Delta x)}{\Delta x}$$



■ $f(x)$ ■ FD ■ True derivative

(forward delta)

$$\frac{f(x + \Delta x) - f(x)}{\Delta x}$$



■ $f(x)$ ■ FD ■ True derivative

Finite Differences and errors



► Clearly, the centre delta visually looks more accurate. Let's prove this using Taylor Series

$$f(x_0 + \Delta x) = f(x_0) + f'(x_0)(\Delta x) + \frac{f''(x_0)}{2!}(\Delta x)^2 + \frac{f'''(x_0)}{3!}(\Delta x)^3 + \frac{f^{(iv)}(x_0)}{4!}(\Delta x)^4 + \dots \quad \boxed{1}$$

(forward delta) $\Rightarrow \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = f'(x_0) + \frac{f''(x_0)}{2!}\Delta x + \frac{f'''(x_0)}{3!}(\Delta x)^2 + \frac{f^{(iv)}(x_0)}{4!}(\Delta x)^3 + \dots = f'(x_0) + \mathcal{O}(\Delta x)$

$$f(x_0 - \Delta x) = f(x_0) + f'(x_0)(-\Delta x) + \frac{f''(x_0)}{2!}(-\Delta x)^2 + \frac{f'''(x_0)}{3!}(-\Delta x)^3 + \frac{f^{(iv)}(x_0)}{4!}(-\Delta x)^4 + \dots \quad \boxed{2}$$

(backward delta) $\Rightarrow \frac{f(x_0 - \Delta x) - f(x_0)}{-\Delta x} = f'(x_0) - \frac{f''(x_0)}{2!}\Delta x + \frac{f'''(x_0)}{3!}(\Delta x)^2 - \frac{f^{(iv)}(x_0)}{4!}(\Delta x)^3 + \dots = f'(x_0) - \mathcal{O}(\Delta x)$

Subtracting 2 from 1,

(centre delta) $\frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x} = f'(x_0) + \frac{f'''(x_0)}{3!}(\Delta x)^2 + \dots = f'(x_0) + \mathcal{O}(\Delta x)^2$

The centre delta has a truncation error of $\mathcal{O}(\Delta x)^2$ vs $\mathcal{O}(\Delta x)$ of both forward and backward delta.

Finite Differences and errors



- We can calculate the second order sensitivity(gamma) using finite difference by adding 1 & 2

$$f(x_0 + \Delta x) + f(x_0 - \Delta x) = 2f(x_0) + 2\frac{f''(x_0)}{2!}(\Delta x)^2 + 2\frac{f^{(iv)}(x_0)}{4!}(\Delta x)^4 + \dots$$

This gives an approximation for 2nd order derivative

$$f''(x_0) \approx \frac{f(x_0 + \Delta x) + f(x_0 - \Delta x) - 2f(x_0)}{(\Delta x)^2}$$